

The Layman's Guide to

Energy AI Hackathon ML Workflow

A Plain-English Explanation of Every Step

For Team Energy Gladiators

What Are We Doing?

We're building a system that predicts how much energy (electricity, diesel fuel, or natural gas) will be used when drilling oil wells using a process called hydraulic fracturing ("fracking").

Think of it like this: You want to predict how much gas your car will use on a road trip. You'd look at the distance, the type of road, the car's efficiency, weather, etc. We're doing the same thing, but for industrial drilling operations.

Step 1: Data Upload & Inspection

What happens: You load two spreadsheets (CSV files) into the system.

- **Training Data:** Information about 1,082 wells where we KNOW how much energy was used
- **Test Data:** Information about 50 wells where we need to PREDICT energy usage

Analogy: The training data is like studying for a test with answer keys. The test data is the actual exam where you apply what you learned.

What to look for:

- How many rows (wells) and columns (features) do you have?
- Are there any missing values (blank cells)?
- What are the column names and data types?

Step 2: Data Cleaning & Imputation

What happens: We fix problems in the data, especially missing values.

"Imputation" means: Filling in blank cells with reasonable guesses.

- **For numbers:** We use the median (middle value) - this is safer than the average because it's not affected by extreme values
- **For categories:** We use the mode (most common value)

Analogy: If a survey respondent left "age" blank, you might fill it with the average age of all respondents. That's imputation.

TIP: Median is often better than average. If most people earn \$50K but one person earns \$10 million, the average salary looks misleadingly high. The median stays realistic.

Step 3: Exploratory Data Analysis (EDA)

What happens: We create charts and statistics to understand our data before building models.

Key Visualizations:

- 1. Histograms (bar charts):** Show how values are distributed. Are most wells clustered around certain energy levels, or spread out evenly?
- 2. Box Plots:** Show the range of values and outliers (unusual data points). The box = middle 50%, whiskers = normal range, dots = outliers.
- 3. Scatter Plots:** Show relationships between two variables. If dots trend upward, the variables are positively correlated.
- 4. Correlation Heatmap:** A grid showing how strongly each variable relates to others. Red = strong positive relationship, Blue = strong negative relationship, White = no relationship.

Analogy: Before cooking a new recipe, you'd examine all your ingredients - their quantities, freshness, how they might combine. EDA is examining your data ingredients.

Key Finding in Our Data: # Stages and # Clusters are the best predictors of energy usage (correlation ~0.25 with CNG). Diesel and CNG have a negative correlation (-0.72) meaning when one is high, the other is low.

Step 4: Feature Engineering

What happens: We create NEW columns from existing data that might help predictions.

Our Engineered Features:

- 1. Time Overrun** = Actual Stage Time - Estimated Stage Time
→ Measures efficiency. Positive = job took longer than planned.
- 2. Total Pumping Time** = # Stages × Actual Stage Time
→ Total work time. Neither stages nor time alone tells the full story.
- 3. Clusters per Stage** = # Clusters / # Stages
→ Work intensity at each stage.

Analogy: Raw data says "drove 60 mph for 2 hours." Engineered feature: "traveled 120 miles." The new feature is often more useful!

Categorical Encoding: Converting text labels (like "Grid", "Diesel") into numbers so the computer can understand them.

Step 5: Model Training

What happens: We teach the computer to predict energy usage by showing it the training data.

We use Random Forest - here's what that means:

Imagine asking 100 different experts to predict energy usage. Each expert sees a slightly different view of the data and makes their own prediction. The final answer is the average of all 100 predictions. That's Random Forest - a "forest" of decision "trees" that vote together.

Configuration Settings:

Number of Trees (100): How many experts vote. More = usually better, but slower.

Max Depth: How detailed each expert's analysis can be. "None" = unlimited depth.

Min Samples Split: Minimum data points needed to make a decision. Prevents overfitting.

Cross-Validation Folds (5): We test the model 5 different ways to make sure it works consistently, not just by luck.

Why 3 Separate Models? Grid wells, Diesel wells, and CNG wells behave differently, so we train a specialist model for each fuel type.

Metrics to Understand:

R² (R-squared): How much of the variation the model explains. 0.80 = explains 80%. Higher is better.

RMSE: Average prediction error in the same units as your target. Lower is better.

MAE: Average absolute error. Easier to interpret than RMSE. Lower is better.

Step 6: Uncertainty Quantification

What happens: We measure how confident we are in each prediction.

No prediction is perfect. The hackathon doesn't just want your best guess - they want to know the RANGE of possible values.

Our Method - Residual Bootstrapping:

1. Calculate "residuals" - the errors from training (Actual - Predicted)
2. For each new prediction, randomly sample 100 past errors
3. Add each sampled error to the prediction to create 100 possible outcomes

Analogy: A weather forecast says "70°F" but also "range: 65-75°F." The range acknowledges uncertainty. We're doing the same for energy predictions.

Step 7: Generate Predictions

What happens: We apply our trained models to the 50 test wells and create the submission file.

The Output File (solution.csv):

- **Masked Well Name:** Identifier for each well
- **Fuel Type:** Grid, Diesel, or CNG
- **Fuel Value:** Our best prediction (point estimate)
- **Real_1 through Real_100:** 100 possible outcomes representing uncertainty

Important: DGB (Dual-fuel) wells produce TWO rows - one for Diesel, one for CNG. That's why 50 wells become 63 rows.

Analogy: Instead of saying "this well will use exactly 50,000 gallons," we say "our best guess is 50,000, but it could realistically be anywhere from 45,000 to 55,000" - and we show 100 examples from that range.

Step 8: Quick Start Guide

What it is: A reference page summarizing how to run through the entire workflow quickly on hackathon day.

This step doesn't do any processing - it's just instructions for rapid execution when you're under time pressure.

Step 9: AI ML Assistant

What it is: A chat interface where you can ask questions about machine learning, get suggestions for features, troubleshoot problems, or adapt the pipeline to different datasets.

This is your innovation differentiator! Other teams build static pipelines. Yours can THINK and ADAPT.

Example questions to ask:

- "What features should I engineer from this data?"
- "Why is my model performing poorly?"
- "Would XGBoost work better than Random Forest here?"
- "How do I interpret this correlation matrix?"

Quick Glossary

Term	Plain English Meaning
Feature	A column in your data - something you measure about each well
Target	What you're trying to predict (Grid kWh, Diesel gal, CNG MMBTU)
Training	Teaching the model using data where you know the answers
Prediction	The model's guess for data where you don't know the answer
Overfitting	When a model memorizes training data but fails on new data
Cross-Validation	Testing the model multiple ways to ensure consistency
Correlation	How strongly two things move together (0=none, 1=perfect)
Residual	The error: Actual value minus Predicted value
Bootstrapping	Randomly sampling from your data to estimate uncertainty
R ² (R-squared)	% of variation explained by the model (0.8 = 80%)
RMSE	Average error size (in same units as target)
Imputation	Filling in missing values with reasonable estimates
Categorical	Data that's a label/category, not a number (e.g., "Grid")
Encoding	Converting categories to numbers so the model understands

Created for Team Energy Gladiators - Good luck at the hackathon!